

Diffusion Policies Preserve Multi-modal Plans: Distilling a Hierarchical Quadrotor Planner for Fast Obstacle Avoidance

Vrishabh Kenkre
Robotics Institute
Carnegie Mellon University
Pittsburgh, PA, USA
vkenkre@cmu.edu

Abstract—Hierarchical planners solve quadrotor obstacle avoidance accurately but at high computational cost: a single replan of our A*-plus-minimum-snap- plus-nonlinear-MPC pipeline costs hundreds of milliseconds, dominated by global search and the model-predictive tracker. We ask whether this pipeline can be distilled into a fast reactive neural-network policy. The key difficulty is multi-modality: when an obstacle lies directly between the drone and its goal, the planner may go left or right with equal validity, so the demonstration data contains two distinct actions for visually similar states. We show that a deterministic multilayer perceptron (MLP) trained on these demonstrations averages the two modes and drives straight into the obstacle, reaching only 2406 ± 2862 mm goal error on decision-point layouts. A diffusion policy trained on the identical data preserves the multi-modal action distribution, samples a single coherent plan per rollout, and reaches the goal at 100 mm—a $24\times$ improvement. As a secondary contribution, we fine-tune the diffusion student with reinforcement learning under a shortest-path reward, which improves aggregate safety metrics—reducing 95th-percentile obstacle field strength on decision-point layouts by 76% (from 1.36 to 0.32) and raising trajectory efficiency from ~ 0.5 to 0.72 —at 18 ms inference. The primary result stands on its own: a learned policy that captures the planner’s discrete homotopy decisions, which a deterministic network of equal or greater capacity cannot.

Index Terms—Diffusion policy, imitation learning, quadrotor, motion planning, reinforcement learning, policy distillation.

I. INTRODUCTION

Quadrotor obstacle avoidance is, in the algorithmic sense, a solved problem. Hierarchical planners that combine grid-based global search, polynomial smoothing, and model-predictive tracking [1]–[3] reliably produce near-optimal collision-free trajectories. The cost is computational: a single replan can take hundreds of milliseconds, dominated by the global search and the constrained nonlinear program solved by the tracker. For deployment on small aerial vehicles with limited onboard compute, a faster controller is desirable.

A natural strategy is *distillation*: train a fast reactive neural network to imitate the slow planner. The planner is the teacher; the network is the student. If successful, the student inherits the planner’s quality while running in microseconds to milliseconds.

The central obstacle to this strategy is *multi-modality*. Consider a single obstacle positioned directly between the drone’s start and its goal. The planner may route left or right; both are valid and roughly equal in cost. Across many such scenes, the demonstration dataset therefore contains two distinct actions—“go left” and “go right”—associated with nearly identical observations. A deterministic network trained to minimize squared error on this data converges to the *mean* action, which points straight ahead, into the obstacle. This is not a capacity problem; it is a structural mismatch between a deterministic function approximator and a multi-modal target distribution.

Diffusion policies [11], which generate actions by iteratively denoising noise conditioned on the observation, represent *distributions* over actions rather than single actions. When the target is bimodal, a diffusion policy samples one of the two modes rather than averaging them. We hypothesize, and confirm, that this property makes diffusion the correct tool for distilling a planner whose demonstrations are multi-modal.

We make three contributions:

- 1) We construct a multi-modal demonstration dataset by solving each scene with multiple randomized homotopy-class plans, and we show empirically that a deterministic MLP collapses on this data while a diffusion policy preserves it—a $24\times$ goal-error improvement on decision-point obstacle layouts.
- 2) We introduce a K -sample inference-time safety filter that exploits the diffusion policy’s stochasticity: K candidate actions are sampled, forward-predicted, and the safest is executed, at negligible runtime cost.
- 3) As a secondary result, we fine-tune the diffusion student with reinforcement learning under a shortest-path reward, improving aggregate safety metrics (95th-percentile field strength down 76% on decision-point layouts) and trajectory efficiency.

The first contribution is the load-bearing one: it isolates policy class, not capacity, as the determining factor in distilling a multi-modal planner.

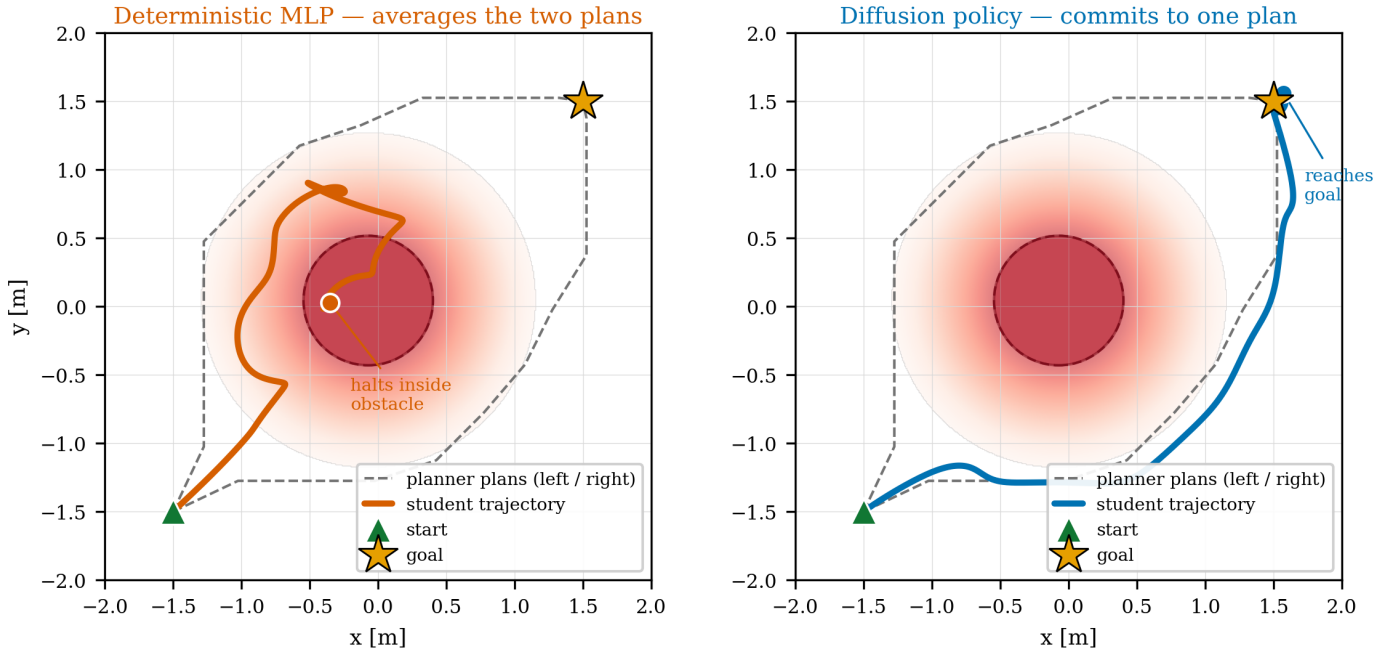


Fig. 1. On a decision-point layout where the planner may route left or right, the deterministic MLP (left) averages the two plans and drives into the obstacle (peak field 1.37); the diffusion policy (right) samples one coherent plan and navigates around it (peak field 0.03). Same dataset, same observation, different policy class.

II. RELATED WORK

Hierarchical quadrotor planning. Minimum-snap trajectory generation [1] and its modern successors MINCO [2] and EGO-Planner [3] produce dynamically feasible references; nonlinear MPC [4], [5] tracks them in closed loop. These methods are accurate but computationally heavy, motivating learned alternatives.

Imitation learning. Behavior cloning [6] is brittle under distribution shift; DAGger [7] and DART [8] mitigate this by expanding the demonstration distribution. All are reactive: they map current observation to action without explicit lookahead, and all assume a unimodal target unless the policy class is chosen otherwise.

Diffusion policies. Diffusion models [9], [10] were adapted to visuomotor control by Chi et al. [11], who demonstrated that the multi-modal expressiveness of diffusion is essential when human demonstrations solve a task in multiple ways. Janner et al. [12] use diffusion for planning directly. Our contribution is to identify planner distillation—rather than human demonstration—as a natural source of multi-modality, and to quantify the deterministic-versus-diffusion gap on a quadrotor task.

Reinforcement fine-tuning of diffusion. DPPO [13] and the method of Black et al. [14] fine-tune diffusion policies with PPO over the denoising chain. Advantage-weighted regression [15] offers a simpler alternative; we adopt it for stability when fine-tuning from a strong behavior-cloned initialization.

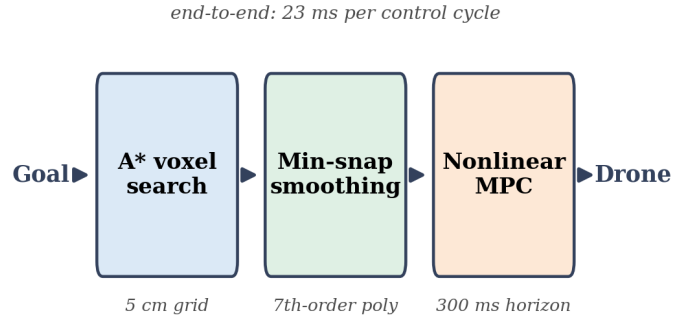


Fig. 2. The hierarchical planner teacher: global A* search over a 5 cm voxel grid, minimum-snap polynomial smoothing, and a 300 ms-horizon nonlinear MPC tracker. End-to-end 23 ms per control cycle.

III. THE HIERARCHICAL PLANNER (TEACHER)

The teacher operates in three stages. **Global search:** 3D A* over a 5 cm voxelized occupancy grid yields a piecewise-linear collision-free path; an Euclidean signed distance field (ESDF) is computed by fast marching. **Smoothing:** the A* waypoints are smoothed by minimizing integral squared snap [1], producing a 7th-order polynomial reference $\mathbf{p}(t)$. **Tracking:** a nonlinear MPC with a 300 ms horizon tracks $\mathbf{p}(t)$, with state the full 12-D quadrotor state and control the four motor commands. Dynamics use the Crazyflie 2.0 model from MuJoCo Menagerie with measured thrust and torque coefficients.

On a suite of 10 random obstacle layouts and 10 hand-

designed decision-point layouts (a single tall obstacle between start and goal), the planner achieves 16 ± 14 mm and 14 ± 3 mm final goal error respectively, with peak obstacle field strength ≤ 0.06 , at 23 ms per control cycle.

IV. DISTILLATION AND THE MULTI-MODALITY PROBLEM

A. Observation and action spaces

The student observes $\mathbf{o}_t \in \mathbb{R}^{24}$, constructed to be *reactive*—it contains no global plan, forcing the policy to infer the homotopy decision from local geometry:

$$\mathbf{o}_t = [\mathbf{s}_t^{(12)}, \phi(\mathbf{p}_t), \nabla\phi(\mathbf{p}_t)^{(3)}, \{\phi(\mathbf{p}_t + \delta_i)\}_{i=1}^8], \quad (1)$$

where $\mathbf{s}_t$ is the 12-D quadrotor state, $\phi(\cdot)$ is the ESDF, $\nabla\phi$ its gradient (the local “away-from-obstacle” direction), and $\{\delta_i\}$ are eight fixed forward probe offsets. We deliberately exclude the planner’s reference from $\mathbf{o}_t$: including it would make every state uniquely identify its rollout and erase the multi-modality by construction. The action $\mathbf{a}_t \in [-1, 1]^4$ is the normalized motor command.

B. Why a deterministic policy must collapse

Let $\pi^*(\mathbf{a} \mid \mathbf{o})$ denote the planner’s (conditional) action distribution. On a decision-point observation $\mathbf{o}$ this distribution is bimodal: mass near $\mathbf{a}_{\text{left}}$ and near $\mathbf{a}_{\text{right}}$. A deterministic network f_θ trained with the squared-error objective $\min_\theta \mathbb{E}_{(\mathbf{o}, \mathbf{a})} \|f_\theta(\mathbf{o}) - \mathbf{a}\|^2$ has the pointwise minimizer

$$f_{\theta^*}(\mathbf{o}) = \mathbb{E}_{\pi^*}[\mathbf{a} \mid \mathbf{o}] \approx \frac{1}{2}(\mathbf{a}_{\text{left}} + \mathbf{a}_{\text{right}}), \quad (2)$$

the conditional mean, which for left/right symmetry points *straight at the obstacle*. This is independent of network capacity: a larger f_θ estimates the same mean more accurately. Escaping the collapse requires a policy class that represents π^* as a distribution rather than its first moment.

C. Constructing multi-modal demonstrations

To make the planner’s multi-modality explicit in the data, we solve each random scene with $K = 3$ randomized A* tiebreaking strategies—generating up to three distinct homotopy-class plans per scene—and each decision-point scene with hand-forced “left” and “right” heuristics. DART noise [8] is injected during tracking: the planner executes $\mathbf{a}_t + \epsilon_t$ with $\epsilon_t \sim \mathcal{N}(0, \Sigma)$ and Σ adapted so the induced state-distribution shift stays below a target total-variation bound, providing off-reference recovery states. The result is ~ 192 k state-action pairs over 658 rollouts. A nearest-neighbor audit (k-NN, $k=10$, radius 0.1 in normalized observation space) confirms that 26% of observation clusters carry an action span exceeding the action range, i.e., they are genuinely multi-modal; a BIC-selected Gaussian mixture flags 7% as multi-modal under a strict criterion.

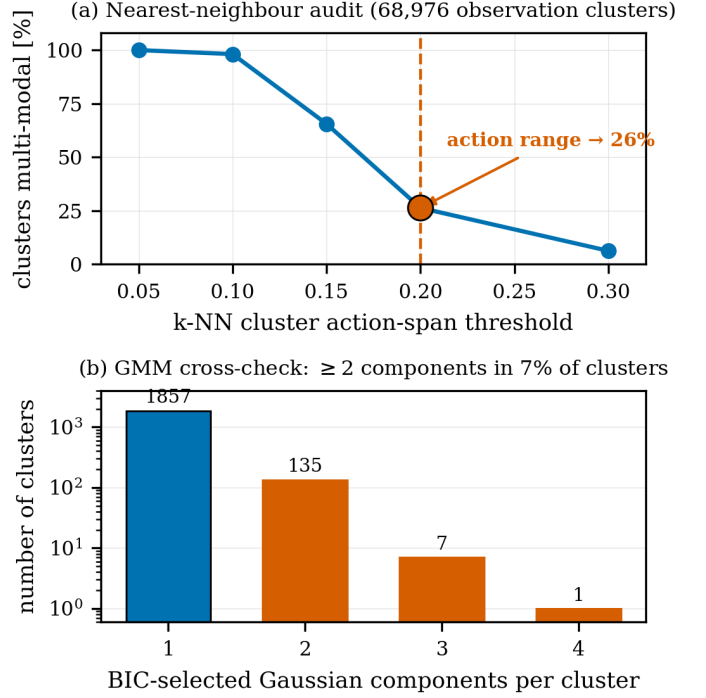


Fig. 3. Action-mode distribution across the demonstration dataset. A nearest-neighbor audit finds 26% of observation clusters carry an action span exceeding the action range, confirming intrinsic multi-modality that a deterministic policy cannot represent.

D. Two students

The **MLP** is a $24 \rightarrow 64 \rightarrow 64 \rightarrow 4$ network (~ 5.7 k parameters) with ReLU activations and a tanh output, trained with DAGger+DART for five outer iterations.

The **diffusion** student is the conditional U-Net of Chi et al. [11] (10.8M parameters), predicting an action chunk $\mathbf{A}_t = (\mathbf{a}_t, \dots, \mathbf{a}_{t+H-1})$ of horizon $H=8$ and executing the first action (receding horizon). Training uses the denoising objective of [9]: a clean action chunk $\mathbf{A}^0$ is noised over $T=100$ steps via $\mathbf{A}^k = \sqrt{\bar{\alpha}_k} \mathbf{A}^0 + \sqrt{1 - \bar{\alpha}_k} \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$, and the network ϵ_θ is trained to predict the added noise conditioned on the observation:

$$\mathcal{L}_{\text{BC}} = \mathbb{E}_{k, \mathbf{A}^0, \epsilon} \|\epsilon - \epsilon_\theta(\mathbf{A}^k, k, \mathbf{o}_t)\|^2. \quad (3)$$

At inference we use 8 DDIM steps [10] for the reverse process, a $25 \times$ reduction from the 100 training steps, yielding 18 ms per action chunk on an RTX 4070. Weights are tracked with an exponential moving average (decay 0.999). Both students consume the same $\mathbf{o}_t$, isolating policy class as the only variable.

E. K-sample safety filter

We exploit the diffusion policy’s stochasticity at inference. At each step we draw K candidate chunks $\{\mathbf{A}_t^{(j)}\}_{j=1}^K \sim \pi_\theta(\cdot \mid \mathbf{o}_t)$, roll each forward under hover-linearized Crazyfly

TABLE I
DISTILLATION ON THE 20-SEED SUITE. “GOAL” IS FINAL POSITION ERROR (MM); “P95_f” IS 95TH-PERCENTILE PEAK OBSTACLE FIELD STRENGTH. DECISION-POINT SEEDS FORCE A LEFT/RIGHT HOMOTOPY CHOICE AND ARE WHERE MULTI-MODALITY DOMINATES.

Controller	Random		Decision-point		Latency
	Goal	P95 _f	Goal	P95 _f	
Planner (teacher)	16	0.05	14	0.06	23 ms
MLP DAgger+DART	846	0.25	2406	1.23	29 μ s
Diffusion ($K=3$)	107	0.62	100	0.98	18 ms

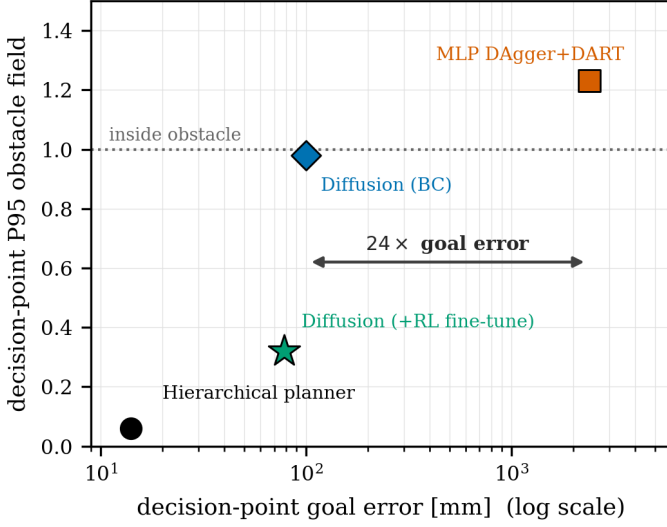


Fig. 4. Goal error versus worst-case obstacle field strength on decision-point layouts. The diffusion policy approaches the planner; the deterministic MLP collapses into the obstacle (field > 1).

dynamics $\hat{\mathbf{p}}_{t+h}^{(j)}$ for $h = 1, \dots, H$, score each by its minimum predicted clearance, and execute the chunk that maximizes it:

$$j^* = \arg \max_j \min_{h \in [1, H]} \phi(\hat{\mathbf{p}}_{t+h}^{(j)}). \quad (4)$$

The K chunks are sampled as a single batched forward pass, so the filter leaves the 18 ms latency essentially unchanged. We use $K=3$.

F. Result

Table I reports the central result. On random scenes both students remain close to the planner. On decision-point scenes the MLP is 24 $\times$ further from the goal and its field strength exceeds 1.0—it drives *into* the obstacle, exactly the mode-averaging failure we predicted. The diffusion student, sampling a single coherent plan per rollout, navigates around the obstacle and reaches the goal. The same dataset, the same observation, a different policy class: that is the contribution.

V. REINFORCEMENT FINE-TUNING WITH A SHORTEST-PATH REWARD

Behavior cloning bounds the student by the teacher’s behavior. To recover trajectory quality and tighten safety, we fine-

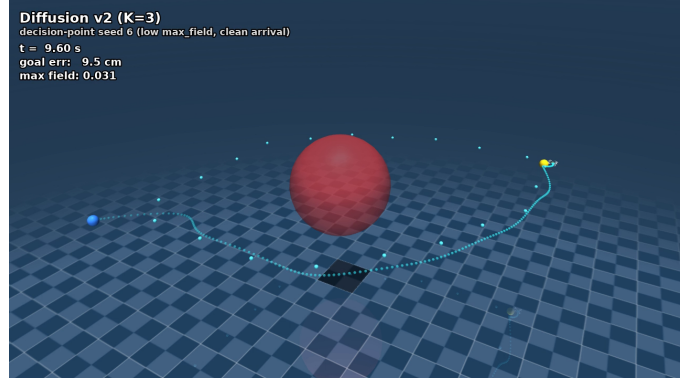


Fig. 5. Diffusion policy on a decision-point layout: it commits to one homotopy class and reaches the goal with peak field strength 0.03.

TABLE II
DIFFUSION BEFORE AND AFTER REINFORCEMENT FINE-TUNING ($K=1$). η_{TRAJ} IS TRAJECTORY EFFICIENCY (STRAIGHT-LINE DISTANCE / PATH LENGTH).

Controller	Random		Decision-point		η_{traj}
	P95 _f	Goal	P95 _f	Goal	
Diffusion (BC)	0.62	93	1.36	69	~ 0.50
+ RL fine-tune	0.19	116	0.32	78	0.72

tune the diffusion student with reinforcement learning under a reward that explicitly favors short, safe, goal-directed paths:

$$r_t = \lambda(d_t^{\text{goal}} - d_{t+1}^{\text{goal}}) - \beta \max(0, 0.15 - \text{esdf}_t) - \eta \|\mathbf{p}_{t+1} - \mathbf{p}_t\| - \gamma \|\mathbf{u}_t\|^2, \quad (5)$$

with a terminal bonus $\kappa \max(0, 0.10 - d_T^{\text{goal}})$. The progress term provides a dense, direction-agnostic gradient to the goal; the path-length term penalizes motion that earns no progress, so the policy prefers the shortest collision-free route whether that passes over or around an obstacle. We use $\lambda=50$, $\beta=8$, $\eta=5$, $\gamma=10^{-3}$, $\kappa=200$.

We optimize by advantage-weighted regression [15] from the behavior-cloned checkpoint, which avoids differentiating returns through the denoising chain. A separate value network V_ψ is fit to rollout returns and warm-started across iterations; generalized advantage estimation gives $\hat{A}_t$. The policy update reweights the BC loss by the exponentiated advantage,

$$\mathcal{L}_{\text{AWR}} = \mathbb{E}_{(\mathbf{o}_t, \mathbf{A}_t, \hat{A}_t)} \left[w_t \|\epsilon - \epsilon_\theta(\mathbf{A}_t^k, k, \mathbf{o}_t)\|^2 \right], \quad w_t = \min(e^{\hat{A}_t/\tau}, c), \quad (6)$$

with temperature $\tau=0.1$ and weight clip $c=5$. We run 15 iterations of 100 rollouts (500-step horizon) at learning rate 10^{-6} , enforcing a relative parameter-drift cap $\|\theta - \theta_{\text{BC}}\| / \|\theta_{\text{BC}}\| < 5\%$ per iteration to prevent the fine-tuning from destabilizing the BC initialization.

Table II shows the effect. Fine-tuning reduces 95th-percentile field strength by 69% on random scenes and 76% on decision-point scenes, and raises trajectory efficiency from ~ 0.5 to 0.72. Goal error rises modestly (≤ 23 mm). We report these as aggregate-metric improvements rather than a

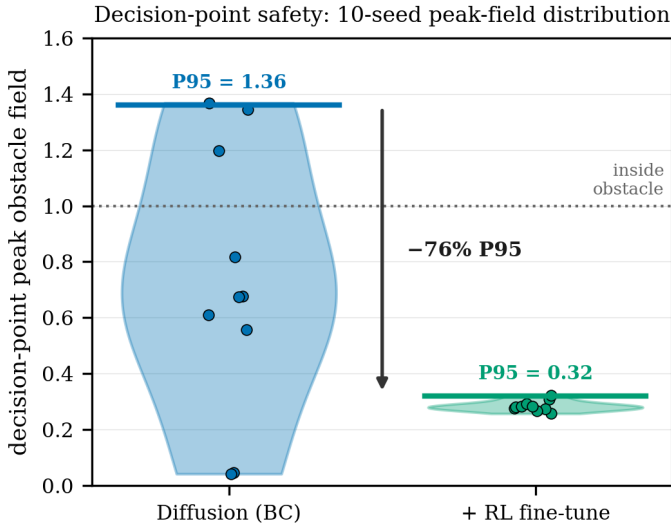


Fig. 6. Reinforcement fine-tuning under the shortest-path reward reduces 95th-percentile obstacle field strength on decision-point layouts by 76%. We report this as an aggregate-metric improvement; see Sec. V for the trajectory-shape caveat.

deployment-ready controller: trajectories under the fine-tuned policy exhibit non-minimal altitude profiles relative to the planner’s reference, particularly on decision-point scenes, and closing this trajectory-shape gap is future work. The fine-tuning serves to demonstrate that the behavior-cloned diffusion student can be improved on explicit objectives; the multi-modality result of Section IV remains the paper’s primary claim.

VI. DISCUSSION

The experiments support a simple, actionable claim: *when a planner’s demonstrations are multi-modal, the policy class matters more than its capacity*. A 5.7k-parameter MLP and a 10.8M-parameter diffusion network are both ample to fit the data; only the diffusion network’s distributional representation avoids mode collapse. Practitioners distilling planners should therefore audit their demonstrations for multi-modality before choosing a deterministic student.

The K -sample safety filter is a cheap, general technique: any stochastic policy can sample multiple candidates and select the safest under a fast forward model, trading a batched GPU evaluation for improved worst-case clearance.

Reinforcement fine-tuning under a carefully shaped reward recovers trajectory quality that behavior cloning alone leaves on the table. The reward must include explicit progress and path-length terms; a safety-only reward yields degenerate evasive behavior. Promising directions include consistency-model distillation [16] to push inference below 5 ms and extension to dynamic obstacles via time-aware distance fields.

VII. CONCLUSION

We distilled the discrete decision-making of a hierarchical quadrotor obstacle-avoidance planner into a fast reactive policy. The decisive design choice is the use of a diffusion policy,

which preserves the planner’s multi-modal homotopy decisions where a deterministic MLP of equal or greater capacity collapses them—a $24\times$ improvement on decision-point obstacle layouts. A K -sample safety filter and shortest-path reinforcement fine-tuning further improve aggregate safety metrics. The primary takeaway is methodological: when distilling a planner whose demonstrations are multi-modal, the policy class, not its capacity, determines success.

REFERENCES

- [1] D. Mellinger and V. Kumar, “Minimum snap trajectory generation and control for quadrotors,” in *IEEE ICRA*, 2011, pp. 2520–2525.
- [2] Z. Wang, X. Zhou, C. Xu, and F. Gao, “Geometrically constrained trajectory optimization for multicopters,” *IEEE Trans. Robotics*, vol. 38, no. 5, pp. 3259–3278, 2022.
- [3] X. Zhou, Z. Wang, H. Ye, C. Xu, and F. Gao, “EGO-Planner: An ESDF-free gradient-based local planner for quadrotors,” *IEEE RA-L*, vol. 6, no. 2, pp. 478–485, 2020.
- [4] M. Kamel et al., “Linear vs nonlinear MPC for trajectory tracking applied to rotary wing micro aerial vehicles,” *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 3463–3469, 2017.
- [5] P. Foehn, A. Romero, and D. Scaramuzza, “Time-optimal planning for quadrotor waypoint flight,” *Science Robotics*, vol. 6, no. 56, 2021.
- [6] D. Pomerleau, “ALVINN: An autonomous land vehicle in a neural network,” in *NeurIPS*, vol. 1, 1989.
- [7] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *AISTATS*, 2011, pp. 627–635.
- [8] M. Laskey et al., “DART: Noise injection for robust imitation learning,” in *CoRL*, 2017, pp. 143–156.
- [9] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *NeurIPS*, vol. 33, 2020.
- [10] Y. Song et al., “Score-based generative modeling through stochastic differential equations,” in *ICLR*, 2021.
- [11] C. Chi et al., “Diffusion policy: Visuomotor policy learning via action diffusion,” *Int. J. Robotics Research*, 2023.
- [12] M. Janner, Y. Du, J. B. Tenenbaum, and S. Levine, “Planning with diffusion for flexible behavior synthesis,” in *ICML*, 2022.
- [13] A. Z. Ren et al., “Diffusion Policy Policy Optimization,” in *NeurIPS*, 2024.
- [14] K. Black et al., “Training diffusion models with reinforcement learning,” in *ICLR*, 2024.
- [15] X. B. Peng, A. Kumar, G. Zhang, and S. Levine, “Advantage-weighted regression,” *arXiv:1910.00177*, 2019.
- [16] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, “Consistency models,” in *ICML*, 2023.